Czech Technical University in Prague

FJFI - Department of Physics

02ELMA - Electricity and Magnetism

Final exam v2

03.06.2025 / 14:00pm - 16:00pm

Name					
Problem	1	2	3	4	Total
Score					

Instructions

1. The time allocated for the exam is 120 minutes.
2. The maximum score for the test is 4 points and each question is worth 1 point.
3. Use of any class materials and calculators are not allowed.
4. Use only pens or non-erasable ink pens for the test; erasable pens are not allowed.
5. Write your answers legibly. Illegible answers will not be scored.
6. Write your answers in the provided space. If you need more space, ask the instructor for additional sheets.

Question 1

Consider a long shell of radius R with a uniform surface charge density σ. Inside this cylindrical shell, coaxially placed, is a long solid cylinder of radius a ($a < R$) with a uniform volume charge density ρ.

(a) Find the electric field in the regions $0 \leqslant r \leq a$, $a \leqslant r \leq R$ and $r \geqslant R$ — 0.6 pts

(b) Determine the potential difference V between the surface of the solid cylinder (radius a) and the cylindrical shell (radius R) — 0.4 pts

Solution:

(a) We use Gauss's Law in cylindrical symmetry:

$$\oint \vec{E} \cdot d\vec{A} = \frac{Q_{\text{enc}}}{\varepsilon_0}$$

where $\vec{E}$ is the electric field, and Q_{enc} is the charge enclosed by a cylindrical Gaussian surface of radius r and length L.

Region I: $0 \leq r \leq a$

In this region, we are inside the solid cylinder. The enclosed charge is:

$$Q_{\text{enc}} = \rho \cdot \pi r^2 L$$

Applying Gauss's Law:

$$E \cdot (2\pi r L) = \frac{\rho \pi r^2 L}{\varepsilon_0} \quad \Rightarrow \quad E = \frac{\rho r}{2\varepsilon_0}$$

Region II: $a \leq r \leq R$

Here, the Gaussian surface encloses the entire solid cylinder:

$$Q_{\text{enc}} = \rho \cdot \pi a^2 L$$

Applying Gauss's Law:

$$E \cdot (2\pi r L) = \frac{\rho \pi a^2 L}{\varepsilon_0} \quad \Rightarrow \quad E = \frac{\rho a^2}{2\varepsilon_0 r}$$

Region III: $r \geq R$ The Gaussian surface now encloses both the solid cylinder and the shell. The total enclosed charge is:

$$Q_{\text{enc}} = \rho \pi a^2 L + \sigma \cdot 2\pi R L$$

Using Gauss's Law:

$$E \cdot (2\pi r L) = \frac{1}{\varepsilon_0} \left(\rho \pi a^2 L + 2\pi R \sigma L\right)$$

$$\Rightarrow E = \frac{1}{2\varepsilon_0 r} \left(\rho a^2 + 2R\sigma\right)$$

(b) The electric potential difference is given by:

$$V(a) - V(R) = -\int_a^R E(r)\, dr$$

In the range $a \leq r \leq R$, we use:

$$E(r) = \frac{\rho a^2}{2\varepsilon_0 r}$$

So:

$$\begin{aligned} V(a) - V(R) &= -\int_a^R \frac{\rho a^2}{2\varepsilon_0 r}\, dr = -\frac{\rho a^2}{2\varepsilon_0} \int_a^R \frac{1}{r}\, dr \\ &= -\frac{\rho a^2}{2\varepsilon_0} \ln\left(\frac{R}{a}\right) \end{aligned}$$

Question 2

Consider a long, straight cylindrical conductor with radius R. The conductor carries a current I uniformly distributed across its cross-sectional area. Surrounding the conductor is a cylindrical shell with an inner radius a (where $a > R$) and outer radius b (where $b > a$). The shell carries a current $-I$ uniformly distributed across its cross-sectional area in the opposite direction.

(a) Find the magnetic field at a distance r from the axis of the conductor for the following regions: $0 \leqslant r \leq R$, $R \leqslant r \leq a$, $a \leqslant r \leq b$ and $r \geqslant b$ — *0.8 pts*

(b) Sketch the magnetic field as a function of r — *0.2 pts*

Solution:

We apply Ampère's Law:

$$\oint \vec{B} \cdot d\vec{\ell} = \mu_0 I_{\text{enc}} \Rightarrow B(2\pi r) = \mu_0 I_{\text{enc}}$$

Region I: $0 \leq r \leq R$:
The current is uniformly distributed, so current enclosed by a circular Amperian loop of radius r is:

$$I_{\text{enc}} = I \cdot \frac{\pi r^2}{\pi R^2} = I \cdot \frac{r^2}{R^2}$$

Hence:

$$B = \frac{\mu_0 I r}{2\pi R^2}$$

Region II: $R \leq r \leq a$:
The entire central conductor is enclosed, and the shell does not contribute here:

$$I_{\text{enc}} = I \Rightarrow B = \frac{\mu_0 I}{2\pi r}$$

Region III: $a \leq r \leq b$:
Now part of the return current in the shell is enclosed. The current density in the shell is:

$$J_{\text{shell}} = \frac{-I}{\pi(b^2 - a^2)}$$

The area of the shell enclosed between a and r is:

$$A = \pi(r^2 - a^2)$$

So the current enclosed from the shell is:

$$I_{\text{shell}}(r) = J_{\text{shell}} \cdot A = -I \cdot \frac{r^2 - a^2}{b^2 - a^2}$$

Total current enclosed:

$$I_{\text{enc}} = I + \left(-I \cdot \frac{r^2 - a^2}{b^2 - a^2}\right) = I\left(1 - \frac{r^2 - a^2}{b^2 - a^2}\right)$$

Then:

$$B = \frac{\mu_0 I}{2\pi r}\left(1 - \frac{r^2 - a^2}{b^2 - a^2}\right)$$

Region IV: $r \geq b$
The total current enclosed is:

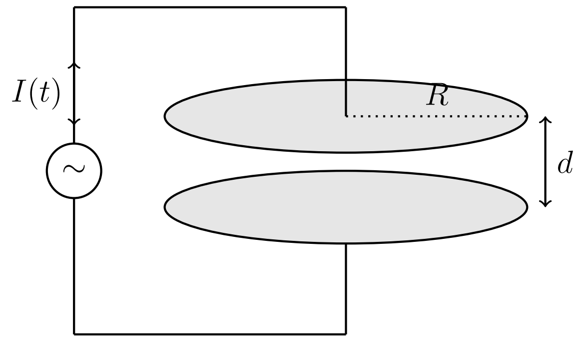
$$I_{\text{enc}} = I + (-I) = 0 \Rightarrow B = 0$$

Question 3

A parallel-plate capacitor with circular plates of radius R and separation d is being charged by a time-varying conduction current

$$I(t) = I_0 \sin(\omega t),$$

through connecting wires, where I_0 is a constant and $\omega \ll 1$ is the angular frequency of the current. The region between the plates is vacuum and $d \ll R$ so that the electric field between the plates is approximately uniform.



(a) Write down the expression for the electric field $E(t)$ between the plates for as a function of instantaneous charge $Q(t)$ and R. — *0.2 pts*

(b) Calculate the charge $Q(t)$ on the plates by using the integral $Q(t) = \int_0^t I(t')dt'$ and express $E(t)$ in terms of I_0, R and ω. — *0.5 pts*

(c) Using $\vec{J}_D = \epsilon_0 \frac{\partial \vec{E}}{\partial t}$, find the magnitude of the displacement current density $J_D(t)$ between the plates. What do you get if you consider the multiplication $J_D \pi R^2$? Would you expect it? Explain it briefly. — *0.2 pts*

(d) Use the Ampère-Maxwell law $\oint \vec{B} \cdot d\vec{l} = \mu_0 \left(I_{\text{enc}} + \epsilon_0 \frac{d\Phi_E}{dt}\right)$ with a circular integration path of radius r (centered on the axis, lying in a plane midway between the plates) to derive an expression for the magnetic field $B(r,t)$ for $r \leq R$. — *0.3 pts*

Solution:

(a) The electric field $E(t)$ between the plates of a parallel-plate capacitor is related to the charge $Q(t)$ on the plates by the formula:

$$\boxed{E(t) = \frac{\sigma}{\epsilon_0} = \frac{Q(t)}{\epsilon_0 A} = \frac{Q(t)}{\epsilon_0 \pi R^2}}$$

(b) The instantaneous charge $Q(t)$ is given by the integral of the current $I(t)$ over time:

$$Q(t) = \int_0^t I(t')dt' = \int_0^t I_0 \sin(\omega t')dt' = \frac{I_0}{\omega}(1 - \cos \omega t)$$

Substituting this expression for $Q(t)$ into the formula for the electric field, we get:

$$\boxed{E(t) = \frac{I_0}{\epsilon_0 \pi R^2 \omega}[1 - \cos(\omega t)]}$$

(c) The displacement current density $J_D(t)$ is related to the time derivative of the electric field by:

$$\boxed{J_D(t) = \epsilon_0 \frac{\partial E(t)}{\partial t} = \frac{I_0}{\pi R^2}\sin(\omega t)}$$

The multiplication $J_D \pi R^2$ gives:

$$J_D \pi R^2 = I_0 \sin(\omega t)$$

This represents the displacement current flowing through the area of the capacitor plates, which is equal to the conduction current $I(t)$ at any instant. It is expected because the displacement current density is defined such that it accounts for the changing electric field in the capacitor, which is necessary to maintain continuity of current in the circuit, especially when the conduction current is not present.
(d) To find the magnetic field $B(r,t)$ for $r \leq R$, let us start with calculating the electric flux for Ampère-Maxwell law:

$$\Phi_E = \int \vec{E} \cdot d\vec{A} = E(t) \cdot \pi r^2$$

Thus, we have:

$$\frac{d\Phi_E}{dt} = \frac{d}{dt}\left(E(t) \cdot \pi r^2\right) = \pi r^2 \frac{dE(t)}{dt}$$

Substituting this into the Ampère-Maxwell law, and considering the conduction current inside the capacitor is zero, we get:

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 \left(0 + \epsilon_0 \pi r^2 \frac{dE(t)}{dt}\right) = \mu_0 \epsilon_0 \pi r^2 \left(\frac{dE(t)}{dt}\right)$$

The left-hand side is $\oint \vec{B} \cdot d\vec{l} = B(r,t) \cdot 2\pi r$, the expression becomes:

$$B 2\pi r = \mu_0 \epsilon_0 \pi r^2 \left(\frac{I_0}{\epsilon_0 \pi R^2} \sin \omega t\right)$$

After simplifying, and rearranging the terms, we find:

$$\boxed{B(r,t) = \frac{\mu_0 I_0}{2\pi R^2} r \sin(\omega t) \text{ for } r \leq R}$$

Question 4

Consider an electromagnetic plane wave propagating in the $+z$ direction in free space. Its electric field is given by

$$\vec{E}(z,t) = E_0 \left[\cos(kz - \omega t)\hat{x} + \alpha \cos(kz - \omega t - \pi/2)\hat{y}\right]$$

where E_0 is a constant, α is a constant concerning the polarization.

(a) Identify the polarization of the wave for $\alpha = 0$ and $\alpha = 1$? — *0.2 pts*

(b) By using the identity $\vec{B} = \frac{1}{c}\hat{z} \times \vec{E}$, find the magnetic field component $\vec{B}(z,t)$. — *0.2 pts*

(c) Compute the instantaneous Poynting vector $\vec{S}(z,t) = \frac{1}{\mu_0}\vec{E} \times \vec{B}$. — *0.3 pts*

(d) Compute the time-averaged Poynting vector $\langle \vec{S} \rangle \equiv \frac{1}{T}\int_0^T \vec{S}(z,t)dt$, where T is the period of the wave. Discuss how the energy flux depends on α. — *0.3 pts*

<u>Given</u>: $\langle \cos^2\theta \rangle = \langle \sin^2\theta \rangle = \frac{1}{2}$

Solution:

(a) For $\alpha = 0$, the electric field is purely in the x-direction, which corresponds to linear polarization along the x-axis. For $\alpha = 1$, the electric field has components in both the x and y directions, which results in circular polarization in the xy-plane, as the two components are equal in magnitude and vary sinusoidally with a phase difference of $\frac{\pi}{2}$.
(b) Calculating the cross product, we have:

$$\vec{B}(z,t) = \frac{1}{c}\hat{z} \times \left[E_0 \cos(kz - \omega t)\hat{x} + \alpha E_0 \sin(kz - \omega t)\hat{y}\right]$$

Using the right-hand rule, we find:

$$\boxed{\vec{B}(z,t) = \frac{1}{c}\left[E_0 \cos(kz - \omega t)\hat{y} - \alpha E_0 \sin(kz - \omega t)\hat{x}\right]}$$

(c) Calculating the cross product, we have:

$$\vec{S}(z,t) = \frac{1}{\mu_0}\left[E_0\left(\cos(kz - \omega t)\hat{x} + \alpha \sin(kz - \omega t)\hat{y}\right) \times \left(\frac{E_0}{c}\cos(kz - \omega t)\hat{y} - \frac{\alpha E_0}{c}\sin(kz - \omega t)\hat{x}\right)\right]$$

Using the properties of the cross product, we find:

$$\boxed{\vec{S}(z,t) = \frac{E_0^2}{\mu_0 c}\left(\cos^2(kz - \omega t) + \alpha \sin^2(kz - \omega t)\right)\hat{z}}$$

(d)The time-average over one period $T = 2\pi/\omega$ gives

$$\left\langle \cos^2\theta + \alpha^2 \sin^2\theta \right\rangle = \frac{1+\alpha^2}{2} \quad \Longrightarrow \quad \boxed{\langle \vec{S} \rangle = \frac{E_0^2}{\mu_0\, c}\,\frac{1+\alpha^2}{2}\,\hat{z}.}$$

Hence the average energy flux scales as $(1+\alpha^2)/2$, giving

- $\alpha = 0$: $\langle S \rangle = E_0^2/(2\mu_0 c)$ (linear),
- $\alpha = 1$: $\langle S \rangle = E_0^2/(\mu_0 c)$ (circular).